CHAPTER 11

Geometric Brownian Motion & Asset Dynamics

Continuous Stochastic Calculus, Itô's Lemma Proof, Lognormal Distributions, and The Volatility Drag Phenomenon

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11.1 From Random Walks to Continuous Price Dynamics

In empirical financial markets, asset prices do not follow deterministic lines; they fluctuate randomly in response to continuous information arrivals. **Geometric Brownian Motion (GBM)** is the foundational continuous-time stochastic process governing modern quantitative finance (including the Black-Scholes-Merton model).

Definition 11.1 (The Stochastic Differential Equation of GBM)

Let asset price at time t be $S_t > 0$, expected annual drift be $\mu \in \blacksquare$, annualized volatility be $\sigma > 0$, and W_t be a standard Wiener process (Brownian Motion) with independent increments $dW_t \sim \blacksquare(0, dt)$.
The relative price return is governed by the Stochastic Differential Equation (SDE):

$$dS_t / S_t = \mu\, dt + \sigma\, dW_t \quad \Rightarrow \quad dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t$$

where $\mu S_t dt$ is the deterministic drift term and $\sigma S_t dW_t$ is the stochastic diffusion shock.

11.2 Analytical Solution via Itô's Lemma

Rigorous Stochastic Calculus Derivation:
In standard calculus, $d(\ln S) = dS / S$. However, in stochastic calculus, Brownian paths have infinite variation, so second-order terms $(dW_t)^2 = dt$ contribute directly to the drift.
Let $f(S) = \ln(S)$. Applying **Itô's Lemma** in one dimension:

- $df = f'(S)\, dS + (1/2)\, f''(S)\, (dS)^2$.
- Derivatives: $f'(S) = 1 / S, \quad f''(S) = -1 / S^2$.
- Second-order term: $(dS)^2 = (\mu S\, dt + \sigma S\, dW)^2 = \sigma^2\, S^2\, dt$.
- Substituting into Itô's expansion:
  $d(\ln S) = (1/S)(\mu S\, dt + \sigma S\, dW) - (1/2)(1/S^2)(\sigma^2\, S^2\, dt) = \mathbf{(\mu - \sigma^2/2)\, dt + \sigma\, dW_t}$.
- Integrating from 0 to t yields the exact analytical trajectory equation:

Theorem 11.1 (Exact Discrete Solution & Volatility Drag)

$$\boldsymbol{S_t = S_0 \cdot \exp[\, (\mu - \sigma^2 / 2) \cdot t + \sigma \cdot W_t\,] = S_0 \cdot \exp[\, (\mu - \sigma^2 / 2) \cdot t + \sigma \sqrt{t} \cdot Z\,]}$$

where $Z \sim \blacksquare(0, 1)$ is a standard normal random variable.
The Volatility Drag Principle: Because of Jensen's inequality and the negative term $-\sigma^2/2$, the **median growth rate** is $\mu - \sigma^2/2$, whereas the arithmetic expected value is $\blacksquare[S_t] = S_0\, e^{\mu t}$. More than 50% of simulated market outcomes strictly underperform the advertised arithmetic mean μ!

11.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: quantifying the volatility drag spread between mean and median, and manually executing a 3-step discretized stochastic trajectory.

Example 11.1: Volatility Drag Quantification over a 10-Year Horizon

Problem Statement: An investor commits capital S_0 **= INR 1,000,000** into a diversified equity fund exhibiting an arithmetic expected return of μ **= 12.00% per annum** and annual volatility of σ **= 20.00%** over a horizon of t **= 10 years**. Calculate the theoretical arithmetic mean, the actual median trajectory, and the exact monetary volatility drag gap.

Step-by-Step Analytical Solution:

Step 1 (Compute Expected Arithmetic Mean Value):

- Formula: $\blacksquare[S_{10}] = S_0 \cdot e^{\mu \cdot t} = 1{,}000{,}000 \times e^{0.12 \times 10} = 1{,}000{,}000 \times e^{1.20}$.
- Since $e^{1.20} \approx 3.3201169$:
- **Theoretical Mean:** $\blacksquare[S_{10}]$ = **INR 3,320,116.92**.

Step 2 (Compute Volatility Drag & Geometric Drift Rate):

- Volatility drag deduction: $\sigma^2 / 2 = (0.20)^2 / 2 = 0.0400 / 2 =$ **0.0200 (2.00% per annum drag!)**.
- Effective geometric median drift: $\mu^* = \mu - \sigma^2/2 = 0.1200 - 0.0200 =$ **0.1000 (10.00% per annum)**.

Step 3 (Compute True Median Wealth Horizon):

- Formula: $\text{Median}(S_{10}) = S_0 \cdot e^{\mu^* \cdot t} = 1{,}000{,}000 \times e^{0.10 \times 10} = 1{,}000{,}000 \times e^{1.00}$.
- Since $e^{1.00} \approx 2.7182818$:
- **Typical Median Outcome:** $\text{Median}(S_{10})$ = **INR 2,718,281.83**.

Step 4 (Volatility Drag Wealth Penalty Analysis):

- Monetary Disparity: 3,320,116.92 – 2,718,281.83 = **INR 601,835.09 gap!**
- *Crucial Insight:* Standard deterministic calculators advertise INR 3.32 Million, but because markets are stochastic, a typical investor will experience only INR 2.72 Million. The probability of falling below the arithmetic mean is $P(Z < \sigma\sqrt{t} / 2) = P(Z < 0.316) \approx$ **62.4% of all investors underperform the average!**

Example 11.2: Discretized 3-Step Stochastic Trajectory by Hand

Problem Statement: Starting from S_0 **= INR 100,000** with drift μ **= 10.0%** and volatility σ **= 15.0%** (Δt = 1 year), simulate a 3-year asset price path by hand using standard normal draws: Z_1 **= +0.65** (bull shock), Z_2 **= –1.10** (bear crash), and Z_3 **= +0.40** (recovery shock).

Step-by-Step Analytical Solution:

Parameters: $\mu = 0.10$, $\sigma = 0.15$, $\Delta t = 1.0$. Volatility drag: $\mu - \sigma^2/2 = 0.10 - (0.0225/2) =$ **0.08875**.
Step equation: $S_k = S_{k-1} \cdot \exp[\, 0.08875 + 0.15 \cdot Z_k \,]$.

Year 1 (Bull Shock, Z_1 = +0.65):
- Exponent: 0.08875 + (0.15 × 0.65) = 0.08875 + 0.0975 = 0.18625.
- Factor: exp(0.18625) ≈ 1.204724.
- **Price Year 1:** S_1 = 100,000 × 1.204724 = **INR 120,472.40** (+20.47%).

Year 2 (Bear Crash, Z_2 = –1.10):
- Exponent: 0.08875 + (0.15 × (–1.10)) = 0.08875 – 0.1650 = –0.07625.
- Factor: exp(–0.07625) ≈ 0.926585.
- **Price Year 2:** S_2 = 120,472.40 × 0.926585 = **INR 111,627.87** (–7.34%).

Year 3 (Recovery Shock, Z_3 = +0.40):
- Exponent: 0.08875 + (0.15 × 0.40) = 0.08875 + 0.0600 = 0.14875.
- Factor: exp(0.14875) ≈ 1.160383.
- **Price Year 3:** S_3 = 111,627.87 × 1.160383 = **INR 129,531.06** (+16.04%).

3-Year Path Summary: INR 100,000 → INR 120,472.40 → INR 111,627.87 → **INR 129,531.06**.
Takeaway: This manual calculation is the exact arithmetic unit executed billions of times across Monte Carlo simulation clusters in quantitative risk management.

11.4 Algorithmic Implementation in Python

Below is the complete implementation from models/monte_carlo.py, providing vectorized Geometric Brownian Motion path generation using NumPy:

Listing: models/monte_carlo.py — Geometric Brownian Motion Path Generator

```
import numpy as np
from typing import Tuple

def simulate_gbm_single_path(
    s0: float, mu: float, sigma: float, time_years: float, steps: int, seed: int = None
) -> Tuple[np.ndarray, np.ndarray]:
    """
    Simulates exact GBM path: S_t = S_0 * exp((mu - 0.5 * sigma^2)*t + sigma * W_t).
    """
    if s0 <= 0.0 or sigma < 0.0 or time_years <= 0.0 or steps <= 0:
        raise ValueError("Invalid parameters for Geometric Brownian Motion.")
    if seed is not None:
        np.random.seed(seed)
    dt = time_years / float(steps)
    time_grid = np.linspace(0.0, time_years, steps + 1)
    drift_step = (mu - 0.5 * (sigma ** 2)) * dt
    diffusion_step = sigma * np.sqrt(dt)
    z = np.random.normal(0.0, 1.0, size=steps)
    log_returns = np.concatenate([[0.0], drift_step + diffusion_step * z])
    price_path = s0 * np.exp(np.cumsum(log_returns))
    return time_grid, price_path
```

11.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Governing SDE	$dS = \mu S\,dt + \sigma S\,dW$	Continuous-time stochastic dynamics of frictionless asset returns
Itô Solution	$S_t = S_0\, e^{(\mu - \sigma^2/2)t + \sigma W_t}$	Closed-form path trajectory driven by Wiener process
Expected Mean	$\blacksquare[S_t] = S_0\, e^{\mu t}$	Arithmetic mean pulled upward by positive lognormal skewness
Typical Median	Median $= S_0\, e^{(\mu - \sigma^2/2)t}$	True geometric median path realized by typical investors
Volatility Drag	$\Delta\mu = -\sigma^2 / 2$	Structural return penalty imposed by variance in continuous time